COMP9123: Data Structures & Algorithms

Abstract Data Types and Algorithm Analysis

Week3

Dr Armin Chitizadeh

Version 1.0

School of Computer Science

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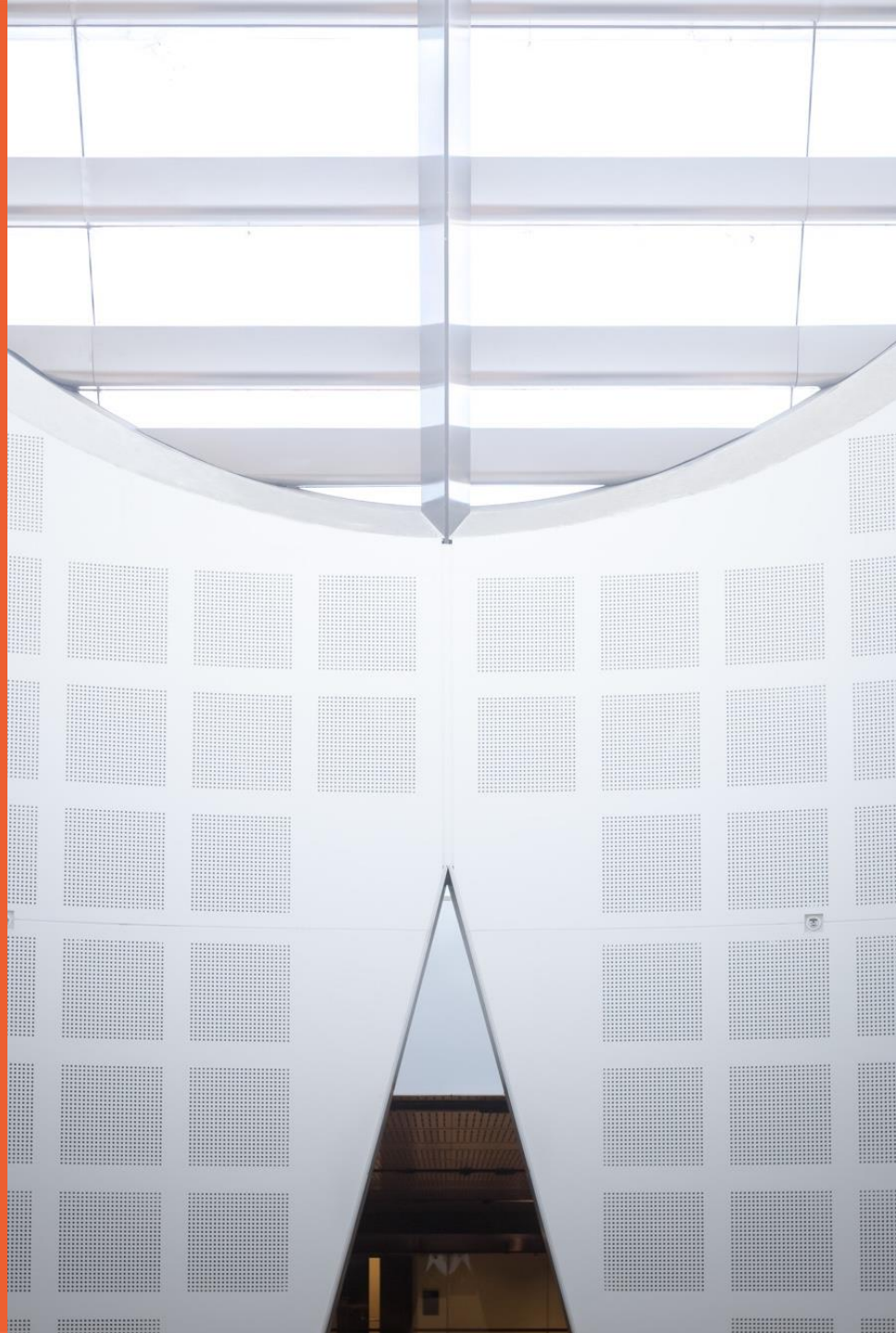
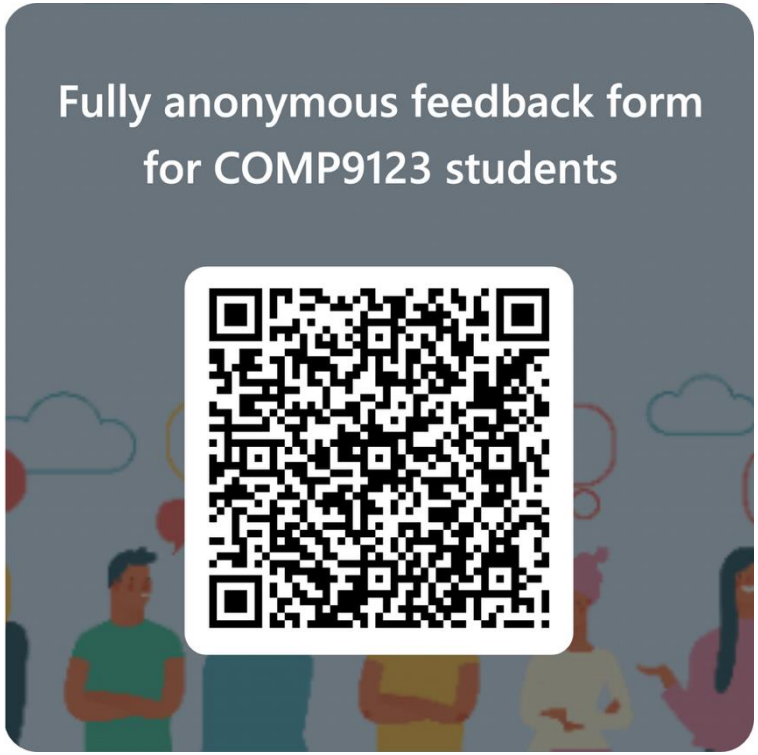


Fully anonymous feedback form for COMP9123 students



Fully anonymous feedback form for COMP9123 students – Fill out form

Week 2 Recap

- We discussed the structure of linked lists and operations
  - Index-based List
  - Positional-based list (Linked List)
- List types
  - Singly linked lists
  - Doubly linked lists
  - Circular lists
- Operations
  - Traversal
  - Insertion
  - Deletion

Any plan for the weekend?

https://www.menti.com/aln4pc6qxh5s

Algorithm Analysis

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Why Do We Need a Performance Measure for Algorithms?

- **Multiple Ways to Solve a Problem** – There are many different algorithms for the same task, and we need a way to evaluate them.

- **Finding the Best Approach** – Algorithm analysis helps compare different solutions to determine which one is the most efficient.

- **Estimating Resource Usage** – It allows us to predict how much time and memory an algorithm will require as input size grows.

- **Ensuring Scalability & Performance** – Helps choose algorithms that work well for both small and large datasets without unnecessary slowdowns.

Early Attempts to Measure Algorithm Efficiency

Measuring Execution Time

- Initially, execution time was used to determine how efficient an algorithm was.
- Developers timed how long an algorithm took to complete a task and compared results.
- While straightforward, this method had significant drawbacks for reliable evaluation.

Why Execution Time is Not a Good Measure?

- **Hardware Dependent** – Performance varies across different CPUs, RAM, and system configurations.

- **Implementation Dependent** – Execution time is affected by programming language, compiler, and system optimizations.

- **Input Size Variation** – Small inputs may run fast, but execution time doesn't predict behavior for large datasets.

- **External Factors** – Background processes and multi-threading can cause inconsistent results.

Real-life example as of 11 March 2026

Hi Armin,

Hope things are well - and beginning of semester is going well.

Please see below from a education reporter. There has been significant software issues with Naplan today - would you have any insight into this/ able to talk about what may have caused it or to prevent future issues?

Thank you

Cheers

The Guardian

Aus

News | Opinion | Sport | Culture | Lifestyle

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Australian education

Naplan resumes after tech issues paused testing on first day for 1.4 million students

Schools told they can begin testing after nationwide problem with pupils logging on to platform

- Follow our Australia news live blog for latest updates
- Get our breaking news email, free app or daily news podcast

https://www.theguardian.com/australia-news/2026/mar/11/naplan-testing-technology-issues

The Need for a Better Measure

Why do we need a new way to analyze algorithms?

- We need a method that is **independent** of hardware and implementation.
- It should focus on how an algorithm scales with input size.
- It must allow us to compare different algorithms objectively.

Introduction to Big-O Notation

What is **Big-O** Notation?

- A mathematical way to describe how the runtime of an algorithm grows with input size (n).
- Focuses on the worst-case scenario to ensure performance reliability.
- Ignores constant factors and lower-order terms. Why?

  (e.g., $O(2n+4n^2) \rightarrow O(n^2)$)

Why is it useful?

- Provides a standardized method for analyzing efficiency.
- Helps choose the best algorithm for large-scale problems.

Understanding Growth Factor in Big-O

- The **growth factor** in time complexity refers to how the runtime of an algorithm increases as the input size (**n**) grows.
- Example:

$$T_{(n)} = \mathrm{n}c_1 + n^2c_2 + n^3c_3 + n^4c_4$$

n=1

$$T_1 = c_1 + c_2 + c_3 + c_4$$

n=10

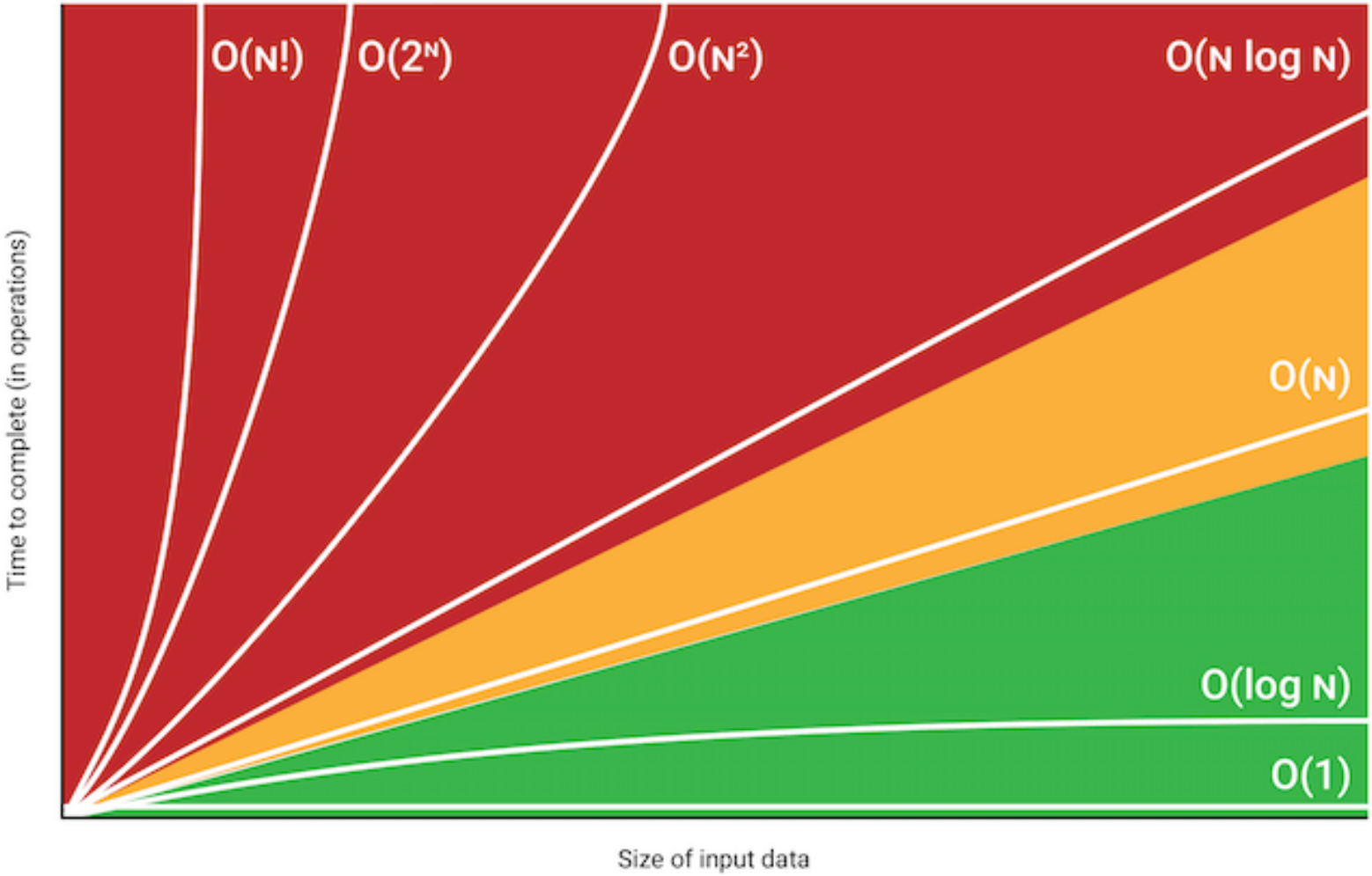
$$T_{10} = 10c_1 + 100c_2 + 1000c_3 + 10000c_4$$

n=100

$$T_{100} = 100c_1 + 10000c_2 + 1000000c_3 + 100000000c_4$$

Big-O = O(n^4)

Big-O Complexity Growth Rates



Why Worst-Case Matters?

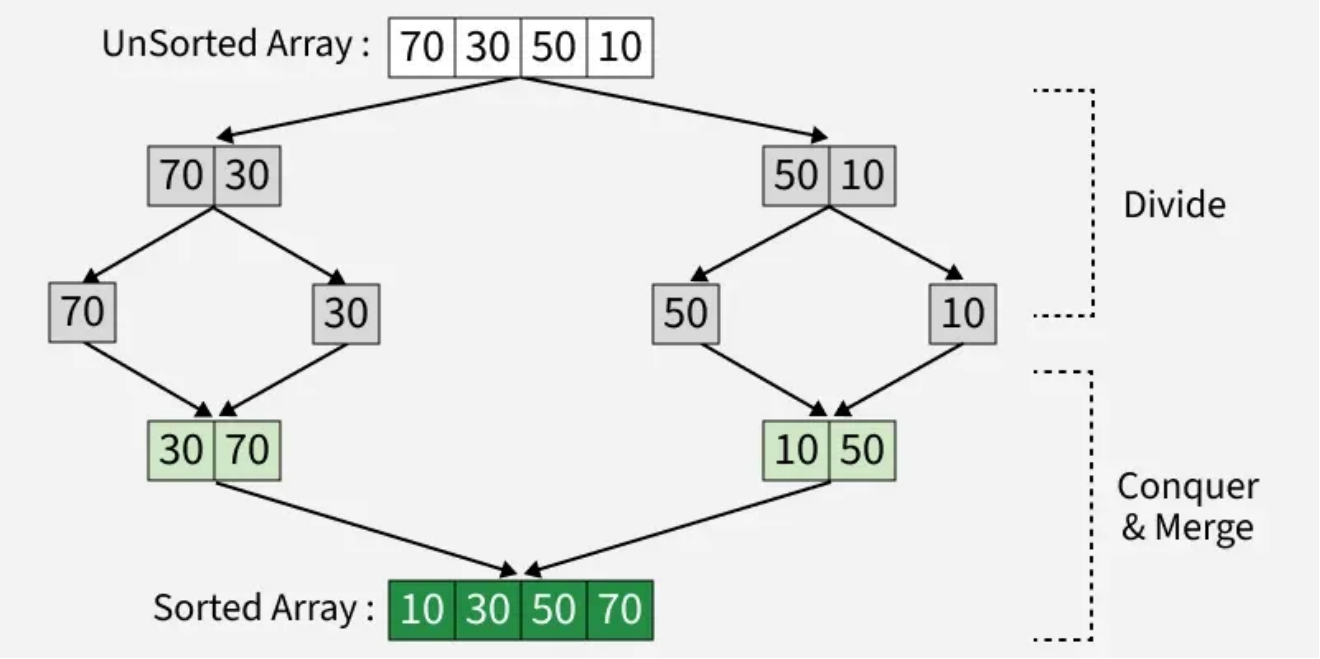
- **Predictability & Reliability**
  – Ensures the algorithm performs within known limits, crucial for real-time and critical applications.

- **Avoids Unexpected Slowdowns**
  – Some algorithms perform well on average but degrade in the worst case
  (e.g., QuickSort: $O(n \log n)$ avg, $O(n^2)$ worst).

- **Helps Choose the Right Algorithm**
  – Algorithms like Merge Sort ($O(n \log n)$) are preferred over Bubble Sort ($O(n^2)$) due to consistent worst-case performance.

- **Optimizes Resource Allocation**
  – Knowing the worst-case complexity helps developers allocate the right amount of computational power, memory, and bandwidth, preventing system crashes and inefficiencies.

Merge Sort - O(n log n)



Bubble Sort - Worst case $O(n^2)$

6 5 3 1 8 7 2 4

Big-O Notation

- Instead of exact times or operations, **Big-O describes growth trends.**
- It tells us the **upper bound (worst-case scenario)** of an algorithm's efficiency.

```
def print_items(n):
    for i in range(n):
        print(i)  # Runs n times
```

O(n) → Linear time complexity

- This gives a **hardware-independent** way to compare algorithms.

Understanding Time & Space Complexity

Time Complexity

- Measures how execution time grows as input size (n) increases.
- Helps analyze algorithm efficiency.
- Example: Searching a name in an unsorted list O(n) vs. binary search in a phonebook O(log n).

Space Complexity

- Measures how much memory an algorithm needs as input size grows.
- Includes variables, recursion, and extra data structures.
- Example: Sorting an array in place O(1) vs. using extra memory for a copy O(n).

Time Complexity Examples

Scanning Items at a Supermarket (O(n)) – Linear Time

Scenario:

- A cashier scans items at checkout, and you have 50 items in your cart.

Approach:

- Each item is scanned one by one into the system. The total time taken grows directly with the number of items.
- If you double the items (100 items), it takes twice as long.

Complexity Analysis:

- The time required increases proportionally with the number of items. Works fine for moderate inputs, but scales linearly.
- Big-O Complexity: O(n) (Good, but not ideal for very large inputs).

Finding a Word in a Physical Dictionary (O(log n)) – Logarithmic Time

Scenario:

- You are looking for the word "Algorithm" in a 1,000-page dictionary.

Approach:

- You don't flip through each page one by one.
- Instead, you open the middle and check:
  - If the word comes before, search the left half.
  - If the word comes after, search the right half.
- You repeat this process until you find the word.

Finding a Word in a Physical Dictionary (O(log n)) – Logarithmic Time

Complexity Analysis:

- Each time, the search space halves (1,000 → 500 → 250 → 125 → 63 → 32 → 16 → 8 → 4 → 2).
- The number of searches needed grows logarithmically with the number of pages.
- Efficiency: Even with a million pages, you'd only need about 20 searches!
- Big-O Complexity: O(log n) (Very efficient for large datasets).

Checking for Duplicate Transactions in a Bank ($O(n^2)$) – Quadratic Time

Scenario:

- A bank needs to check for duplicate transactions in a list of 1,000 payments.

Approach:

- The system compares each transaction with every other transaction to see if they match.
- This requires nested loops:
  - The first loop picks a transaction.
  - The second loop checks all other transactions for a duplicate.

Checking for Duplicate Transactions in a Bank ($O(n^2)$) – Quadratic Time

Complexity Analysis:

- If there are 1,000 transactions, the system performs 1,000 × 1,000 = 1,000,000 comparisons.
- If transactions double to 2,000, comparisons become 4,000,000—this scales poorly!
- Big-O Complexity: $O(n^2)$ (Becomes too slow for large datasets).

Common Time Complexity Examples with Real-World Analogies

Complexity	Example	Real-World Analogy
O(1) (Constant)	Accessing arr[i]	Finding a book by its shelf number
O(log n) (Logarithmic)	Binary search	Looking for a word in a dictionary
O(n) (Linear)	Looping through an array	Checking every page in a book
O(n log n) (Linearithmic)	Merge Sort	Efficiently organizing pizza orders
O(n^2) (Quadratic)	Bubble Sort	Pairwise comparisons in a tournament
O(2^n) (Exponential)	Recursive Fibonacci	Brute-force password cracking

Space Complexity Examples

Storing Customer Orders in a Database (O(n)) – Linear Space

Scenario:

- A restaurant keeps track of customer orders in a system.

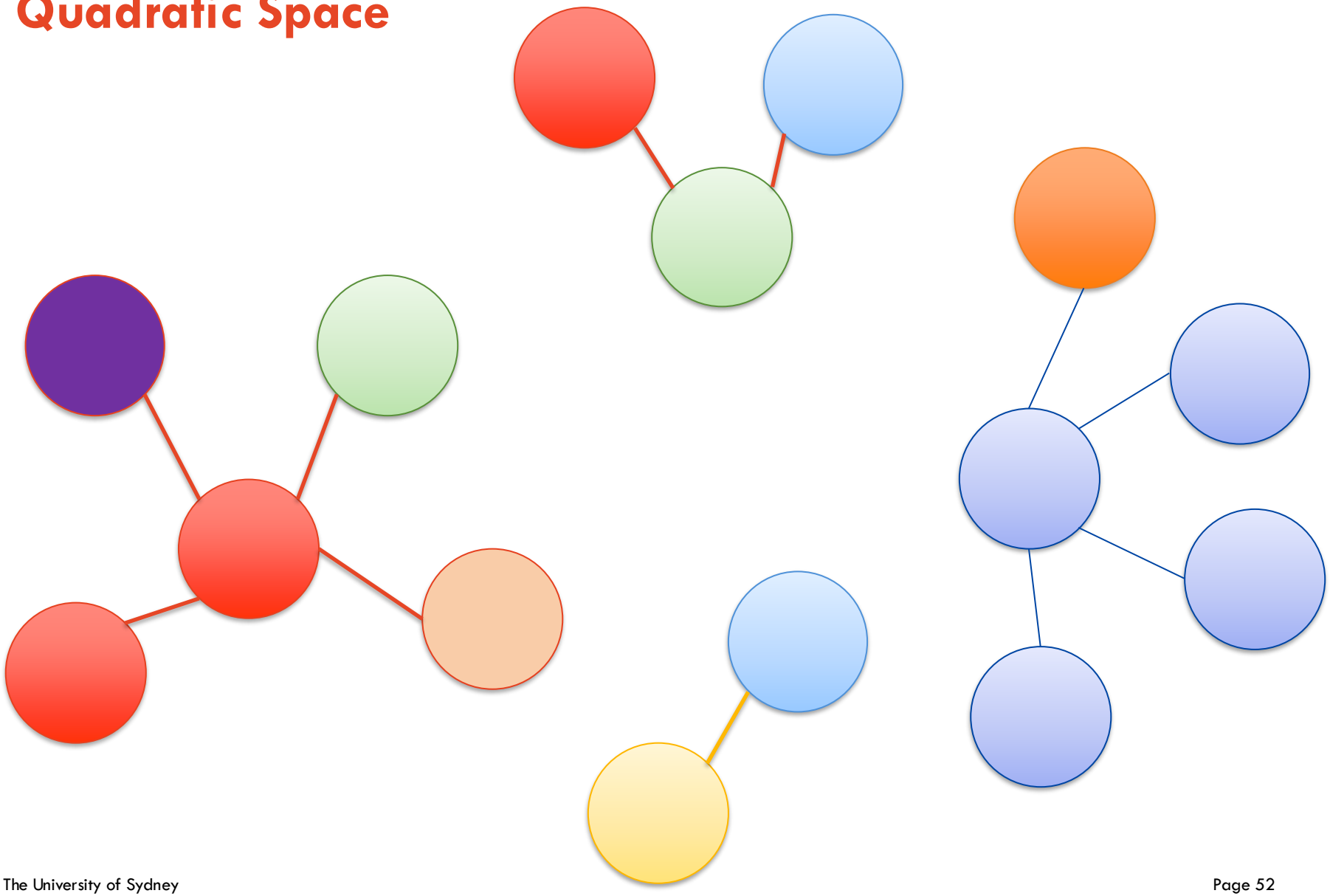
Approach:

- Each new order is stored in a list.
- As the number of customers increases, memory usage grows proportionally.

Complexity Analysis:

- The storage required increases linearly with the number of orders.
- Big-O Complexity: O(n) (Reasonable for moderate inputs but can grow significantly).

Social Network Friend Recommendations ($O(n^2)$) – Quadratic Space



Social Network Friend Recommendations ($O(n^2)$) – Quadratic Space

Scenario:

- A social media platform suggests friends based on mutual connections.

Approach:

- The system stores a list of each user's friends and compares mutual friends.
- If a platform has 1,000 users, the system may store 1,000 × 1,000 connections.

Complexity Analysis:

- As users increase, memory grows quadratically.
- Big-O Complexity: $O(n^2)$ (Efficient for small networks but challenging for large-scale social media).

Generating All Possible Wi-Fi Passwords ($O(2^n)$) – Exponential Space

Scenario:

- A brute-force password cracking system generates all possible Wi-Fi passwords.

Approach:

- If a password is 8 characters long, the system generates $2^8 = 256$ possible passwords.
  - You could assume that we are considering a password after it's been converted into binary code.
- If the password is 16 characters, it stores $2^{16} = 65{,}536$ possibilities.

Complexity Analysis:

- Each new character doubles the memory required.
- Big-O Complexity: $O(2^n)$ (Unmanageable for long passwords).

Feature Subset Selection in Machine Learning ($O(2^n)$) – Exponential Space

Selecting the optimal subset of features in Machine Learning

Example: Electricity usage prediction

- Population size
- Average household size
- Weather temperature
- Industrial activity level
- Commercial building usage
- Number of households
- Air conditioning usage
- Public transportation electrification
- Energy efficiency of buildings
- Economic activity level

Common Space Complexity Examples with Real-World Analogies

Complexity	Example	Real-World Analogy
O(1) (Constant)	Storing the top 10 highest scores in a game	A leaderboard that always keeps only the top 10 scores, no matter how many players join
O(n) (Linear)	Storing user browsing history	A browser history where every new website visited takes up additional space
O(n^2) (Quadratic)	Storing all possible direct flight connections between cities	A flight route map where every city is directly connected to every other city, requiring storage for each pair
O(2^n) (Exponential)	Generating all possible chess game scenarios	A chess tournament bracket where every new move doubles the number of possible game outcomes

Menti!

- https://www.menti.com/althhmoim881

Break!

- Menti

Stacks

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Stacks and Queues

These ADTs are restricted forms of List, where insertions and removals happen only in particular locations:

- stacks follow last-in-first-out (LIFO)
- queues follow first-in-first-out (FIFO)

So why should we care about a less general data structures?

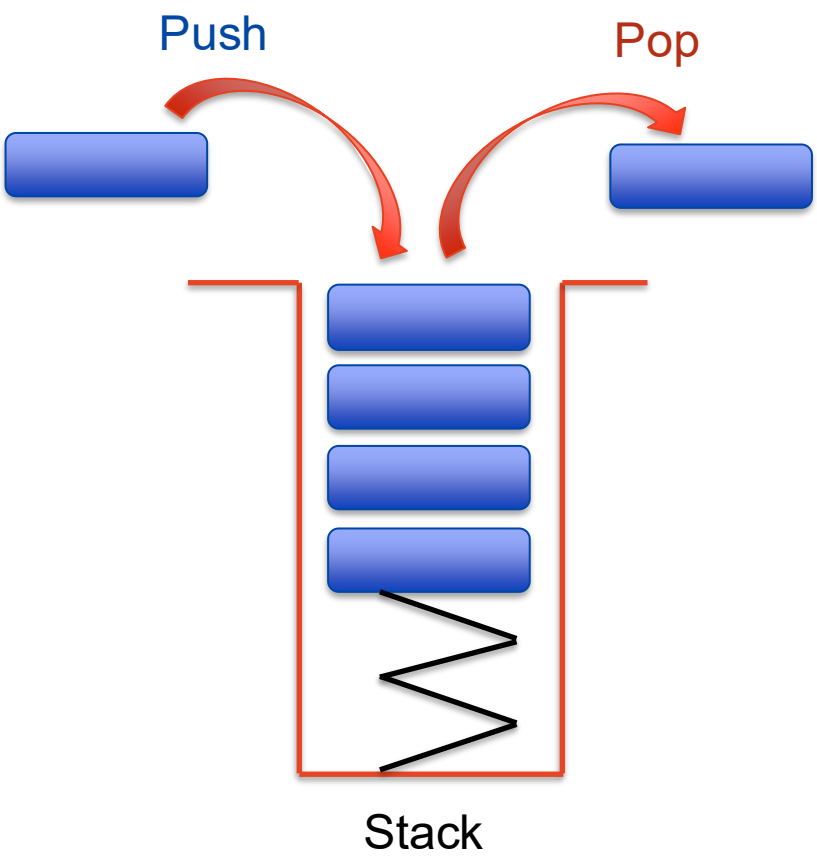
- operations and names are part of computing culture
- numerous applications
- simpler/more efficient implementations than Lists

What is a Stack?

- A stack (sometimes called a "push-down stack") is an ordered collection of items where the addition of new items and the removal of existing items always takes place at the same end.
- This end is commonly referred to as the "top"
- Stack principle: Last In First Out (LIFO) which means the last element inserted is the first one to be removed
- Example: Which is the first element to be picked up?

Stack Example

- A common model of a stack is a plate or coin stacker.
- Plates are "**pushed**" onto the top and "**popped**" off from the top
- Last-In-First-Out (LIFO)



Stack Operations

Main stack operations:

- **push**(e): inserts an element, e
- **pop**(): removes and returns the last inserted element

Auxiliary stack operations:

- top(): returns the last inserted element without removing it
- size(): returns the number of elements stored
- isEmpty(): indicates whether no elements are stored

Stack Operation Examples

Operation	Returns	Stack
push(5)	-	[5]
push(3)	-	[5, 3]
size()	2	[5, 3]
pop()	3	[5]
isEmpty()	False	[5]
pop()	5	[]
isEmpty()	True	[]
push(7)	-	[7]
push(9)	-	[7, 9]
top()	9	[7, 9]
push(4)	-	[7, 9, 4]
pop()	4	[7, 9]

Question

Given an initial stack with elements [6, 9, 3], perform the following operations in sequence:

- push(5)
- pop()
- top()
- push(8)
- isEmpty()
- pop()
- top()

What is the resulting stack?

Question

Given an initial stack with elements [6, 9, 3], perform the following operations in sequence:

- push(5)
- pop()
- top()
- push(8)
- isEmpty()
- pop()
- top()

What is the resulting stack? [6, 9, 3]

Question

Given an initial stack with elements [6, 9, 3], perform the following operations in sequence:

- Initial stack: [6, 9, 3]
- push(5): [6, 9, 3, 5]
- pop(): [6, 9, 3] (5 is removed)
- top(): 3 (stack remains [6, 9, 3])
- push(8): [6, 9, 3, 8]
- isEmpty(): False (stack is not empty)
- pop(): [6, 9, 3] (8 is removed)
- top(): 3 (stack remains [6, 9, 3])

Resulting stack? [6, 9, 3]

Applications of Stacks

Direct applications

- Keep track of a history that allows undoing such as Web browser history or undo sequence in a text editor
- Chain of method calls in a language supporting recursion
- Parentheses checker-examine a file to see if its braces {} and other operators are matching

Indirect applications

- Auxiliary data structure for algorithms
- Component of other data structures

Method Stacks

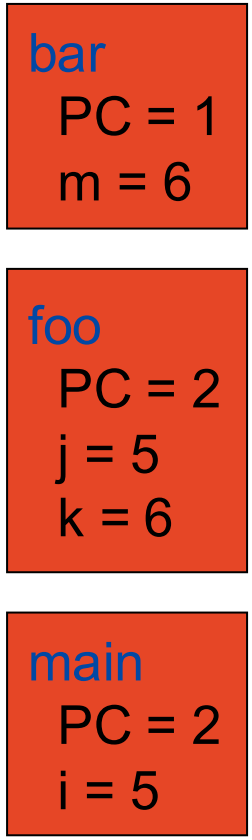
The runtime environment keeps track of the chain of active methods with a stack, thus allowing recursion.

When a method is called, the system pushes on the stack a frame containing

- Local variables and return value
- Program counter

When a method ends, we pop its frame and pass control to the method on top

```
def main()
    i = 5;
    foo(i);

def foo(j)
    k = j+1;
    bar(k);

def bar(m)
    ...
```



Balanced Parentheses

- When analyzing arithmetic expressions, it is important to determine whether an expression is balanced with respect to parentheses

  (a + b * (c / (d – e))) + (d / e)

- The problem is further complicated if braces or brackets are used in conjunction with parentheses
- The solution is to use stacks!

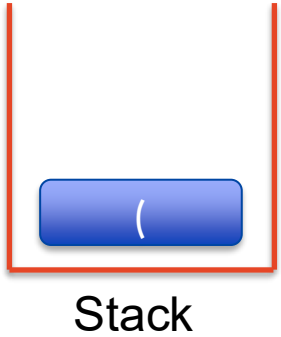
Steps to Check for Balanced Parentheses

- Initialize an empty stack.
- Iterate through each character in the expression.
  - If the character is an **opening bracket** (`(` , `{` , `[`), push it onto the stack
  - If the character is a **closing bracket** (`)` , `}` , `]`):
    - Check if the stack is empty. If yes, return **false** (unbalanced).
    - Otherwise, **pop** the top element from the stack.
    - Check if the popped opening bracket **matches** the current closing bracket. If not, return **false** (unbalanced).
- After iteration, check the stack:
  - If the stack is **empty**, return **true** (balanced).
  - If not, return **false** (unbalanced).

Balanced Parentheses (Example 1)

Expression: (w * [x + y] / z)

0	1	2	3	4	5	6	7	8	9	10
(	w	*	[	x	+	y	]	/	z	)

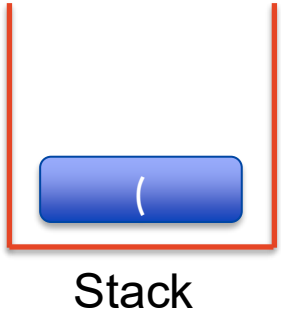


Balanced : true
Index : 0

Balanced Parentheses (Example 1)

Expression: (w * [x + y] / z)

0	1	2	3	4	5	6	7	8	9	10
(	w	*	[	x	+	y	]	/	z	)

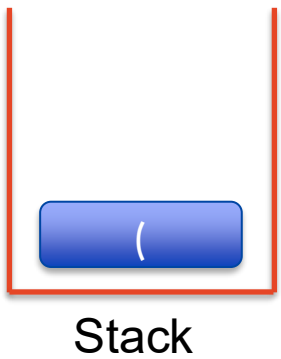


Balanced : true
Index : 1

Balanced Parentheses (Example 1)

Expression: (w * [x + y] / z)

0	1	2	3	4	5	6	7	8	9	10
(	w	*	[	x	+	y	]	/	z	)

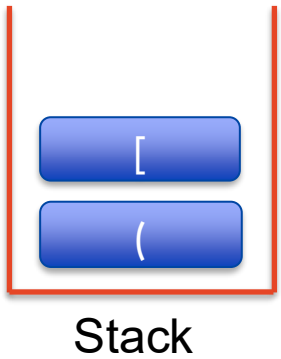


Balanced : true
Index : 2

Balanced Parentheses (Example 1)

Expression: (w * [x + y] / z)

0	1	2	3	4	5	6	7	8	9	10
(	w	*	[	x	+	y	]	/	z	)

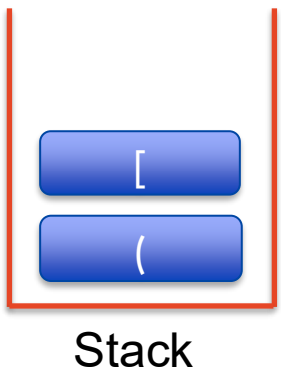


Balanced : true
Index : 3

Balanced Parentheses (Example 1)

Expression: (w * [x + y] / z)

0	1	2	3	4	5	6	7	8	9	10
(	w	*	[	x	+	y	]	/	z	)

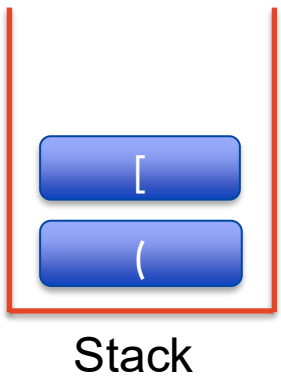


Balanced : true
Index : 4

Balanced Parentheses (Example 1)

Expression: (w * [x + y] / z)

0	1	2	3	4	5	6	7	8	9	10
(	w	*	[	x	+	y	]	/	z	)

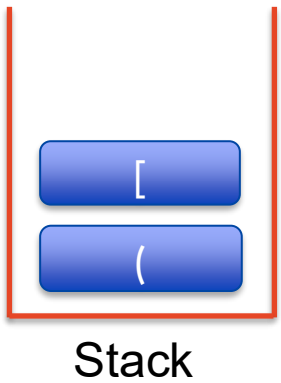


Balanced : true
Index : 5

Balanced Parentheses (Example 1)

Expression: (w * [x + y] / z)

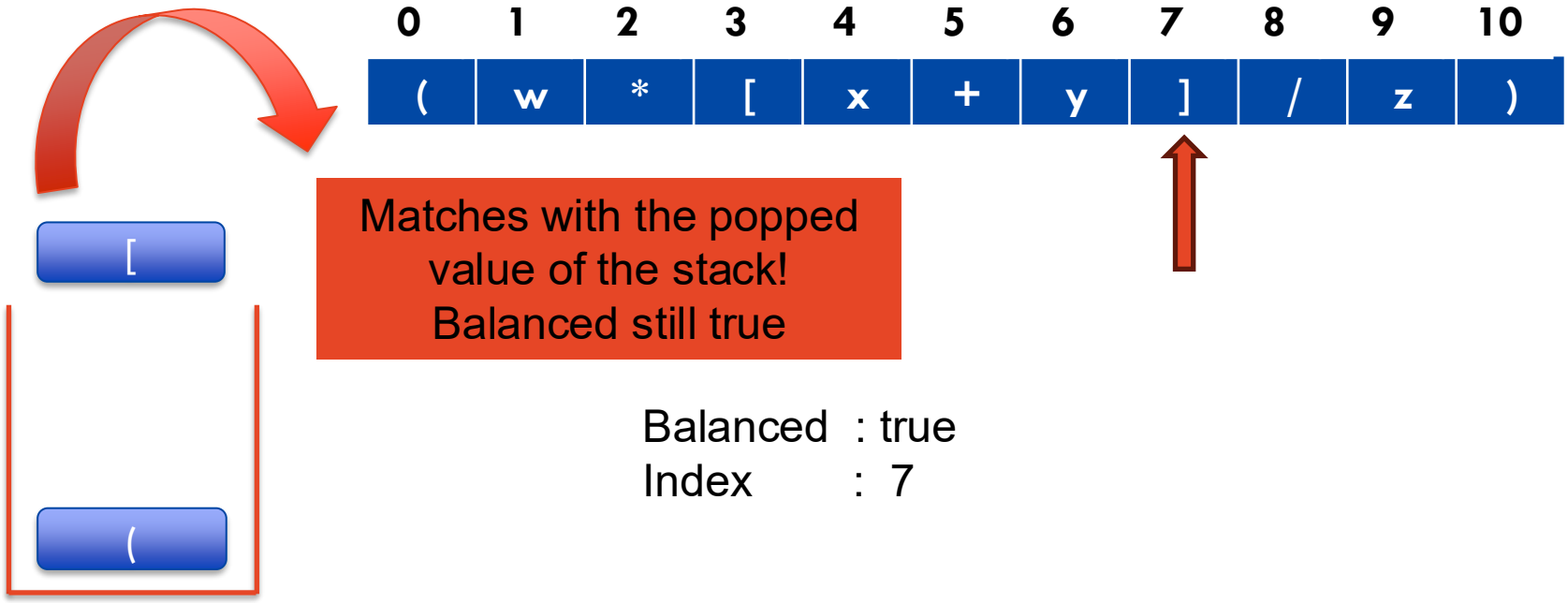
0	1	2	3	4	5	6	7	8	9	10
(	w	*	[	x	+	y	]	/	z	)



Balanced : true
Index : 6

Balanced Parentheses (Example 1)

Expression: (w * [x + y] / z)



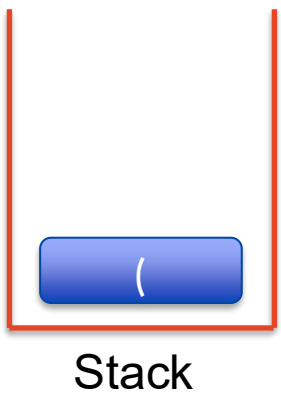
Matches with the popped value of the stack!
Balanced still true

Balanced : true
Index : 7

Balanced Parentheses (Example 1)

Expression: (w * [x + y] / z)

0	1	2	3	4	5	6	7	8	9	10
(	w	*	[	x	+	y	]	/	z	)

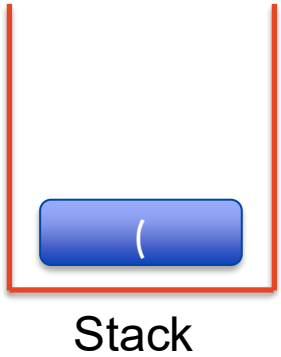


Balanced : true
Index : 8

Balanced Parentheses (Example 1)

Expression: (w * [x + y] / z)

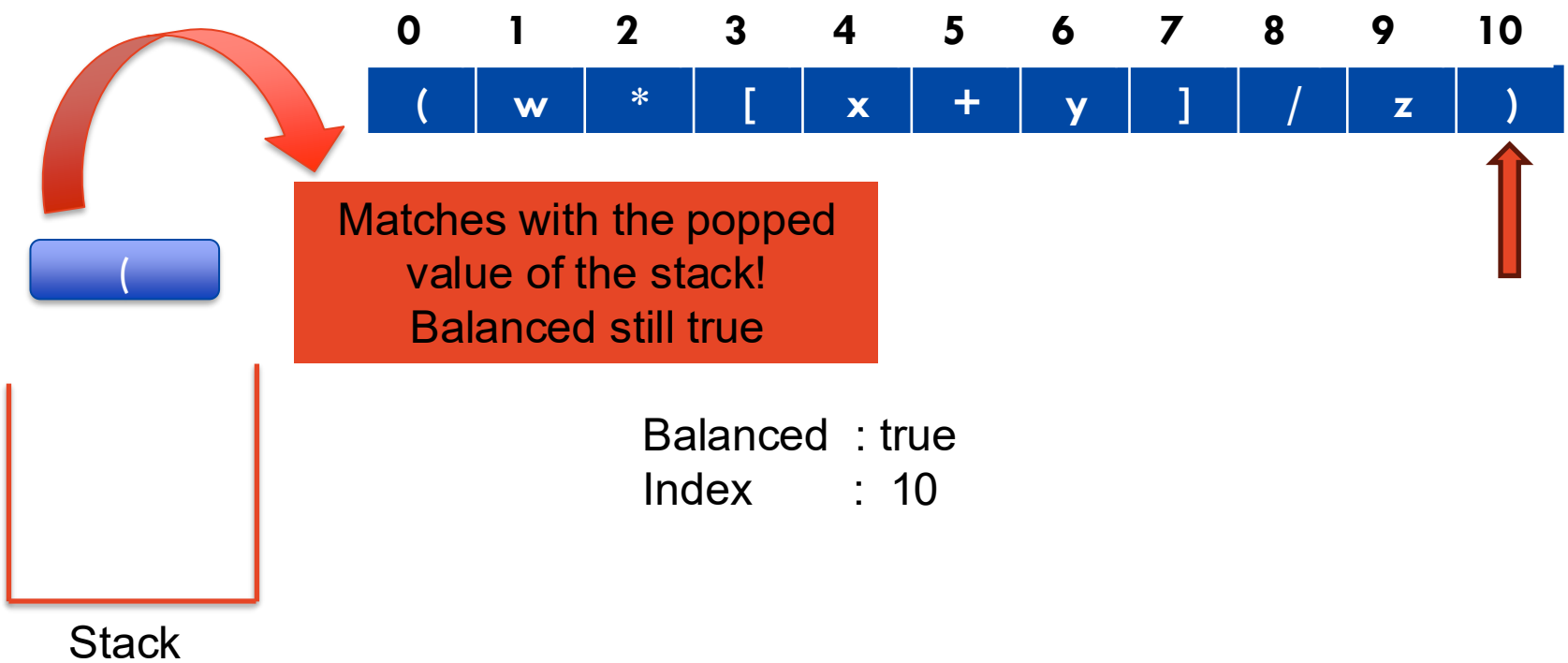
0	1	2	3	4	5	6	7	8	9	10
(	w	*	[	x	+	y	]	/	z	)



Balanced : true
Index : 9

Balanced Parentheses (Example 1)

Expression: (w * [x + y] / z)

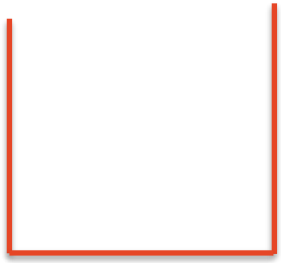


Balanced : true
Index : 10

Balanced Parentheses (Example 1)

Expression: (w * [x + y] / z)

0	1	2	3	4	5	6	7	8	9	10
(	w	*	[	x	+	y	]	/	z	)



Stack

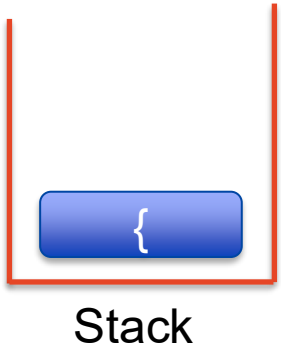
Balanced : true
Index : 10

Since the stack is empty at the end, the expression is **balanced**

Balanced Parentheses (Example 2)

Expression: { [x + y)] - z }

0	1	2	3	4	5	6	7	8	9
{	[	x	+	y	)	]	-	z	}

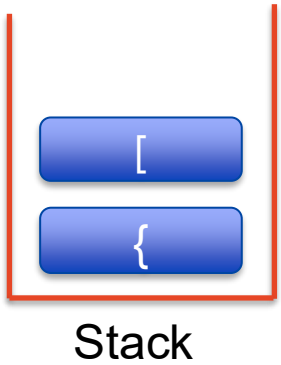


Balanced : true
Index : 0

Balanced Parentheses (Example 2)

Expression: { [x + y)] - z }

0	1	2	3	4	5	6	7	8	9
{	[	x	+	y	)	]	-	z	}

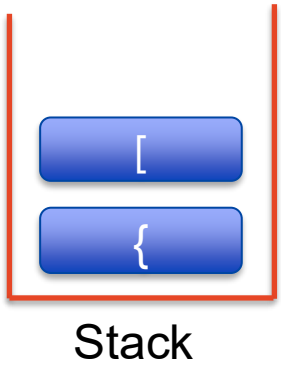


Balanced : true
Index : 1

Balanced Parentheses (Example 2)

Expression: { [x + y)] - z }

0	1	2	3	4	5	6	7	8	9
{	[	x	+	y	)	]	-	z	}

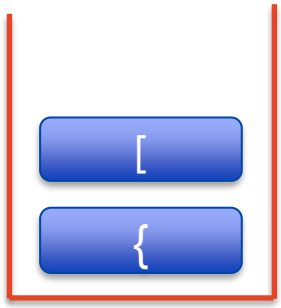


Balanced : true
Index : 2

Balanced Parentheses (Example 2)

Expression: { [x + y)] - z }

0	1	2	3	4	5	6	7	8	9
{	[	x	+	y	)	]	-	z	}



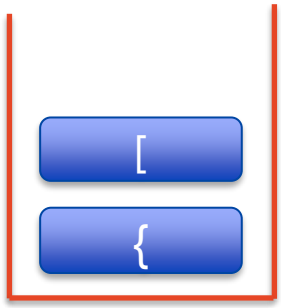
Stack

Balanced : true
Index : 3

Balanced Parentheses (Example 2)

Expression: { [x + y)] - z }

0	1	2	3	4	5	6	7	8	9
{	[	x	+	y	)	]	-	z	}

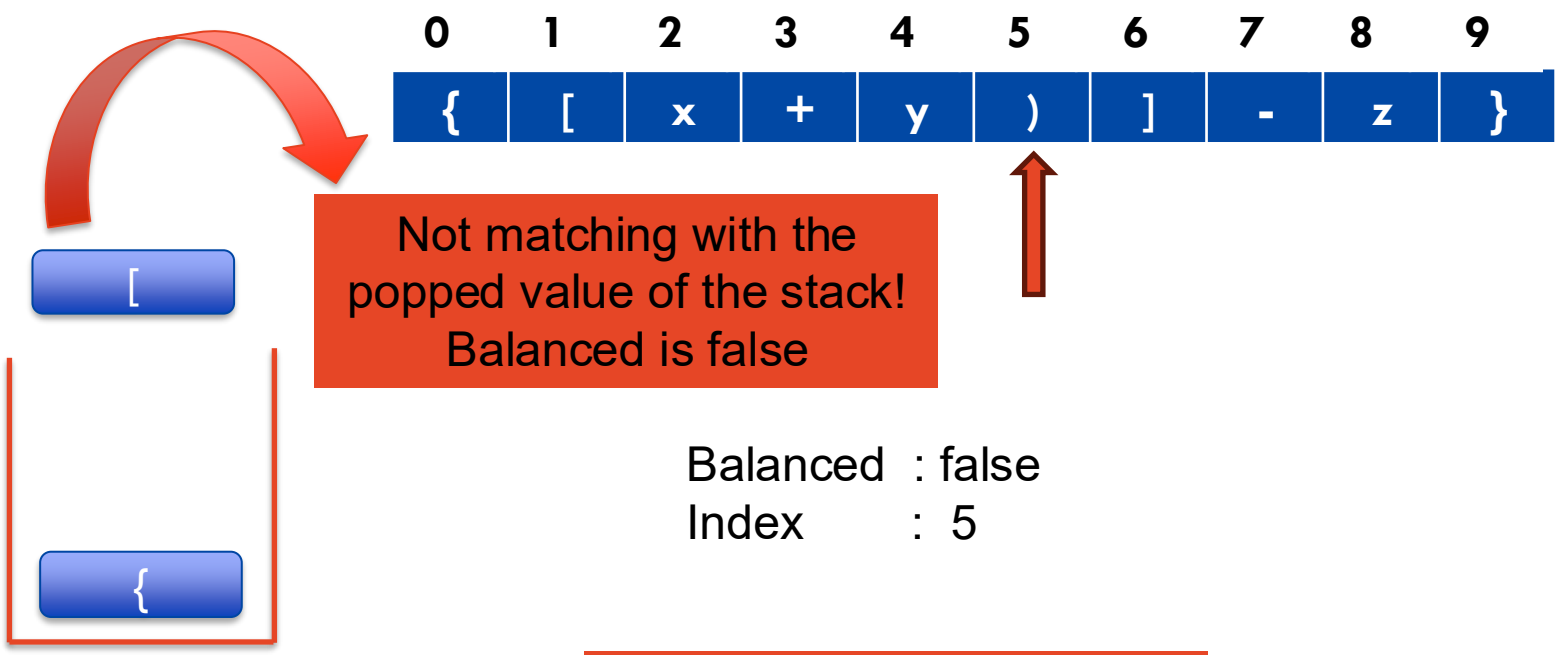


Stack

Balanced : true
Index : 4

Balanced Parentheses (Example 2)

Expression: { [x + y)] - z }



Not matching with the popped value of the stack! Balanced is false

Balanced : false
Index : 5

Since the Balanced is false, the expression is **unbalanced**

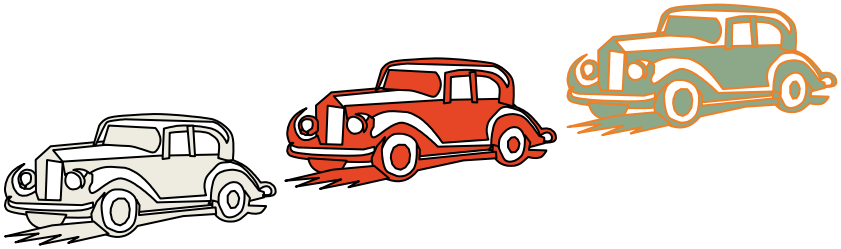
Balanced Parentheses – Python Implementation

```python
def is_balanced(statement):
    stack = Stack()
    pairs = {')': '(', '}': '{', ']': '['}

    for character in statement:
        if character in pairs.values():
            stack.push(character)
        elif character in pairs.keys():
            if stack.isEmpty():
                return False
            elif stack.pop() != pairs[character]:
                return False

    return stack.isEmpty()
```

Queues

THE UNIVERSITY OF SYDNEY

Queue ADT



Main queue operations:

- enqueue(e): inserts an element, e, at the end of the queue
- dequeue(): removes and returns element at the front of the queue

Auxiliary queue operations:

- first(): returns the element at the front without removing it
- size(): returns the number of elements stored
- isEmpty(): indicates whether no elements are stored

Queue Example

Operation	Output	Queue
enqueue(5)	-	(5)
enqueue(3)	-	(5, 3)
dequeue()	5	(3)
enqueue(7)	-	(3, 7)
dequeue()	3	(7)
first()	7	(7)
dequeue()	7	()
isEmpty()	true	()
enqueue(9)	-	(9)
enqueue(7)	-	(9, 7)
size()	2	(9, 7)
enqueue(3)	-	(9, 7, 3)
enqueue(5)	-	(9, 7, 3, 5)
dequeue()	9	(7, 3, 5)

Question

Given an initial queue with elements [6, 9, 3], perform the following operations in sequence:

- enqueue(5)
- dequeue()
- first()
- enqueue(8)
- size()
- dequeue()
- first()

What is the resulting queue?

Question

Given an initial queue with elements [6, 9, 3], perform the following operations in sequence:

- enqueue(5)
- dequeue()
- first()
- enqueue(8)
- size()
- dequeue()
- first()

What is the resulting queue? [3, 5, 8]

Question

Given an initial queue with elements [6, 9, 3], perform the following operations in sequence:

- Initial queue: [6, 9, 3]
- enqueue(5): [6, 9, 3, 5]
- dequeue(): [9, 3, 5] (6 is removed)
- first(): 9 (queue remains [9, 3, 5])
- enqueue(8): [9, 3, 5, 8]
- size(): 4 (queue has 4 elements)
- dequeue(): [3, 5, 8] (9 is removed)
- first(): 3 (queue remains [3, 5, 8])

Resulting queue? [3, 5, 8]

Queue applications

Buffering packets in streams, e.g., video or audio

Direct applications

- Waiting lists
- Access to shared resources (e.g., printer)
- Multiprogramming

Indirect applications

- Auxiliary data structure for algorithms
- Component of other data structures

Queue Application: Ticket Counter

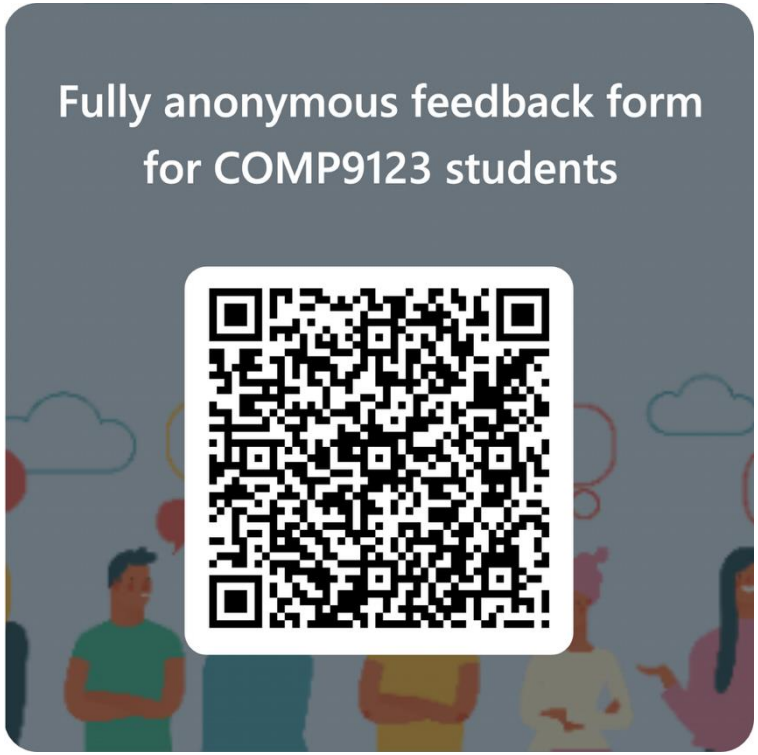
Imagine a queue at a ticket counter where people stand in line to buy tickets. The first person in line gets served first, following the **FIFO (First in, First Out)** principle.

Operations in the queue:

1. Enqueue (Add to queue) : A person joins the end of the queue
2. Dequeue (Remove from queue) : The person at the front gets served and leaves the queue

Updates:

Fully anonymous feedback form for COMP9123 students



Fully anonymous feedback form for COMP9123 students – Fill out form